Switches

* its devices connect/disconnect between two Point.

* type

① Mechincal switch
↳ need humanPower to change between state to another → Push, switch

② electriCal switch
↳ need electricty to change between state to another → transister, Relay

* Mechincal Connection

① External PullUP
② Internal PullPull
③ External Pull down

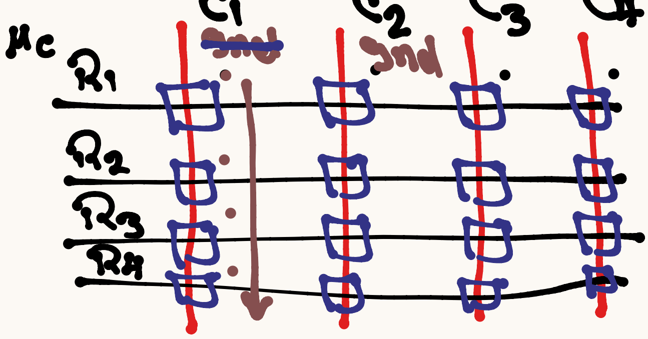
Control b/ 1Pin

if You need to Connect 16 Push Button
↳ 16 Pin from Mc

* key Pad

↳ Mechincal switch (Push Button)
↳ Connection b/ Matrix
↳ to Minimize No of Pin

Key Pad Circuit

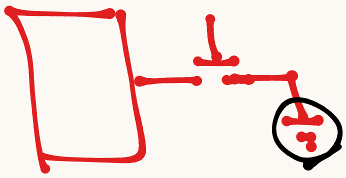


~~C1~~, C2, C3, C4 → High 5v

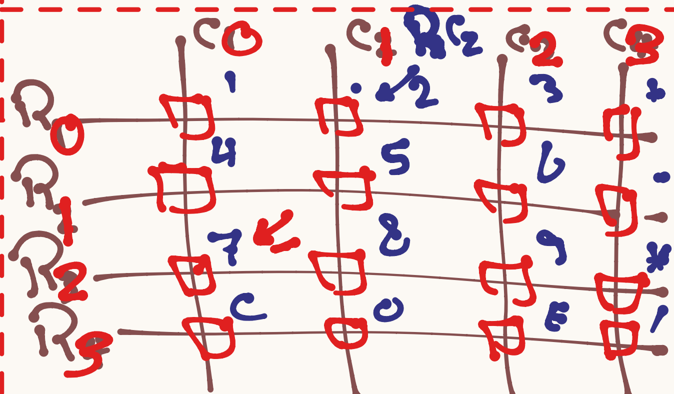
~~gnd~~ gnd R1 R2 R3 R4

Vcc

* Normal Button (Internal PullUP)



Pressed = 0
NotPress = 1



4 4
R3 C1

```
arry [R][C] =
{
R0 → { '1', '2', '3', '+' },
{ '4', '5', '6', '-' },
{ '7', '8', '9', '*' },
{ 'C', '0', 'E', '/' }
};
```

* R0 R1 R2 R3 → MC
INPUT
↳ Internal pullup.

* C0 , C1 C2 C3 → MC
↳ output
↳ Inial High

KPd Init

* loop (Colouns.)
↳ // Colour → low
↳ loop (Rows)
↳ Read R
↳ Return value From array.
↳ // Colour → High.

Gnd

* de bounce

* busy bug.